Problem

Design a high-pass filter with a high-frequency gain of 5 and a corner frequency of 2 kHz. Use a 50-nF capacitor in your design.

Step-by-step solution

Step 1 of 3

The gain of the high pass filter is,

$$\frac{-R_f}{R_i} = 5$$

Capacitor value is,

$$C_i = 50 \text{ nF}$$

Corner frequency is,

$$f_c = 2 \text{ kHz}$$

Calculate the corner frequency in rad/s.

$$\begin{aligned}\omega_c &= 2\pi f_c \\ &= 2\pi\left(2\times10^3\right) \\ &= 12.566 \text{ krad/s}\end{aligned}$$

Step 2 of 3

For a high pass filter, the expression for high pass filer in terms of capacitor and resistor is,

$$\omega_c = \frac{1}{R_i C_i}$$

Substitute 12.566 krad/s for ω_c, and 50 nF for C_i in the above equation.

$$12.566\times10^3 = \frac{1}{R_i\left(50\times10^{-9}\right)}$$

$$\begin{aligned}R_i &= \frac{1}{\left(12.566\times10^3\right)\left(50\times10^{-9}\right)} \\ &= \frac{1}{628.32\times10^{-6}} \\ &= 1591.6\ \Omega \\ &\cong 1600\ \Omega\end{aligned}$$

Therefore, the value of resistor, R_i is $\boxed{1600\ \Omega}$.

Step 3 of 3

The gain of the filter is,

$$\frac{R_f}{R_i} = 5$$

Substitute $1600\ \Omega$ for R_i in the above equation.

$$\frac{R_f}{1600\ \Omega} = 5$$

$$\begin{aligned} R_f &= (5)(1600\ \Omega) \\ &= 8\ \text{k}\Omega \end{aligned}$$

Therefore, the value of the resistor R_f is $\boxed{8\ \text{k}\Omega}$.